5·12·2023(M)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: July 2023 –October 2023 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 54 Programme: Honours in Data Science and Analytics.	Class: LY-Honours	Semester: VII (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer Engineering
Course Code: 116h54C701	Name of the Course: Advanced Machine Learning	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	What are the advantages of CNN (Convolutional Neural network) over ANN (Artificial neural network).	5
ii)	What is the benefit of using Momentum in Gradient Descent optimization?	5
iii)	Why Seq2Seq model is called as Conditional language model? Justify your answer.	5
iv)	Why Sigmoid or Tanh is not preferred to be used as the activation function in the hidden layer of the neural network?	5
v)	You are designing a deep learning system to detect driver fatigue in cars. It is crucial that your model detects fatigue, to prevent any accidents. Which of the following is the most appropriate evaluation metric: Accuracy, Precision, Recall, and Loss Value? Explain your choice.	5
vi)	Describe in short five phases of NLP	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Consider a neural network with three output neurons for a multiclass classification task. The network produces raw scores [2.0, 1.0, 0.5] as its output for a given input. Calculate the probabilities for each class using the softmax activation function. Explain how these probabilities can be interpreted in the context of classification.	5
ii)	Consider the Objective Function: $J(\theta) = (\theta - 4)^2$, $\theta_0 = 3$ (random initial value) $\alpha = 0.1$ (learning rate). Calculate values of θ_1 & θ_2 using gradient descent method.	5
	OR	
Q2 A	Compare Deep Neural Networks (DNNs) with Shallow Neural Networks based on following parameters: i. Architecture Complexity ii. Feature Representation iii. Training challenges iv. Generalization Capability v. Computational efficiency	10

Q 2 B	Solve any **One**	**10**
i)	Name and explain any Four hyper-parameters used for training a deep neural network?	10
ii)	Explain k fold cross validation technique with example.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain forward and backward propagation in simple RNN with suitable mathematical equations.	10
ii)	Describe the role of the input gate, forget gate, and output gate in an LSTM network. How do these gates contribute to the network's ability to capture long-term dependencies?	10
iii)	Let us consider a Convolutional Neural Network having three different convolutional layers in its architecture as – Layer-1: Filter Size – 3 X 3, Number of Filters – 10, Stride – 1, Padding – 0 Layer-2: Filter Size – 5 X 5, Number of Filters – 20, Stride – 2, Padding – 0 Layer-3: Filter Size – 5 X5 , Number of Filters – 40, Stride – 2, Padding – 0 If we give the input image to the network of dimension 39 X 39 X 3, then determine the dimension of the vector after passing through a fully connected layer in the architecture.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Describe the architecture of a typical Seq2Seq model. Include the roles of the encoder and decoder, and explain how the model processes variable-length sequences.	10
ii)	Discuss one real-world application where Seq2Seq models are commonly employed. How does the architecture of Seq2Seq models make them suitable for this application?	10
iii)	Discuss the training process of the CBOW model in Word2Vec word embedding model. Include the roles of the input and output layers, the softmax function, and how word vectors are updated during training.	10

Que. No.	Question	Max. Marks
Q5	Write a short note on any **Four**	**20**
i)	Text pre-processing techniques (Any Two)	5
ii)	Adam Optimizer	5
iii)	Sentiment analysis	5
iv)	One hot encoding	5
v)	Cross-entropy loss	5
vi)	Different types of pooling in CNN.	5